

Name **Suggested Solutions**



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination
Higher 2

Class

Reg Number

17S

Calculator Model / No.

Chemistry

9729/04

Paper 4

11 September 2018

2 hours 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name, register number and class in the space provided above.

Give details of the practical shift and laboratory where appropriate, in the boxes provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use correction fluid.

Answer **all** questions in the space provided on the Question Paper.

The use of approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

Qualitative Analysis Notes are provided on Pages **23** and **24**.

At the end of the examination, fasten all your work accurately together.

The number of marks is given in brackets [] at the end of each question or part question.

Shift

Laboratory

Examiner's Use

Q1

/ 21

Q2

/ 19

Q3

/ 15

Total

/ 55

This document consists of **24** printed pages (including this cover page).

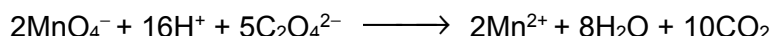
- 1** In this experiment, you are to determine the number of water of crystallisation (value of x) in an iron(II) salt, $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot x \text{H}_2\text{O}$ by titration with potassium manganate(VII).

You are provided with the following reagents.

- FA 1** is potassium manganate(VII), KMnO_4 solution, of an approximate concentration 0.02 mol dm^{-3}
FA 2 is $0.0780 \text{ mol dm}^{-3}$ ethanedioate ions $\text{C}_2\text{O}_4^{2-}$
FA 3 is a solution of the iron(II) salt containing 38.4 g dm^{-3} of $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot x \text{H}_2\text{O}$.
FA 4 is dilute sulfuric acid

Before using the provided solution of manganate(VII) ions, it is important to accurately determine its concentration. The exact concentration of **FA 1** can be determined with the given **FA 2** solution.

In an acidic medium, ethanedioate ions $\text{C}_2\text{O}_4^{2-}$ reacts with manganate(VII) ions MnO_4^- to form Mn^{2+} and CO_2 gas.



As this reaction is very slow initially, MnO_4^- should only be added to a pre-heated $\text{C}_2\text{O}_4^{2-}$ solution in the presence of excess dilute acid. Subsequently, as the reaction proceeds, the rate of reaction will speed up and hence eliminating the need to continue heating.

Method

(a) (i) Standardisation of FA 1 solution

1. Fill the burette with **FA 1**.
2. Using a pipette, transfer 10.0 cm^3 of **FA 2** into the conical flask.
3. Using an appropriate measuring cylinder, transfer 25 cm^3 of **FA 4** to the same conical flask.
4. Heat this solution to about 65°C using the Bunsen burner.
5. **Turn off your Bunsen burner.**

Caution: Use the heat-resistant foam to handle the neck of the heated flask.

6. Run **FA 1** from the burette into this flask until a **permanent** pale pink colour is obtained.

Initially, the colour of the **FA 1** will take some time to disappear. Subsequently, as the reaction proceeds, the rate of reaction will speed up

7. Record your titration results in the space provided on page 3. Make certain that your recorded results show the precision of your working.
8. Repeat steps 1 to 7 as necessary until consistent results are obtained.

Results

	1	2
Final burette reading / cm ³	15.10	15.10
Initial burette reading / cm ³	0.30	29.90
Volume of FA 1 / cm ³	14.80	14.80

✓

✓

- (ii) From your titrations, obtain a suitable volume of FA 1 to be used in your calculations. Show clearly how you obtained this volume.

[2]

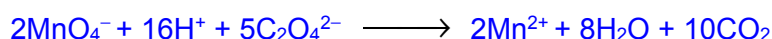
$$\text{Average volume of FA 1 used} = (14.80 + 14.80) \div 2 = \underline{14.80 \text{ cm}^3} \text{ (2 d.p.)}$$

volume of FA 1 =

- (iii) Determine the concentration of MnO₄⁻ in FA 1.

[1]

$$\text{Amount of C}_2\text{O}_4^{2-} \text{ used (in 10.0 cm}^3\text{)} = 0.0780 \times \frac{10.0}{1000} = 7.80 \times 10^{-4} \text{ mol}$$



$$\text{Amount of MnO}_4^- \text{ reacted (in 14.80 cm}^3\text{)} = \frac{2}{5} \times 7.80 \times 10^{-4} = 3.12 \times 10^{-4} \text{ mol}$$

$$[\text{MnO}_4^-] = 3.12 \times 10^{-4} \div \frac{14.80}{1000} = \underline{0.0211 \text{ mol dm}^{-3}} \text{ (3 s.f.)}$$

Concentration of MnO₄⁻ in FA 1 =

M1	M2	M3

- (iv) For the titration in (a)(i) between ethanedioate ions C₂O₄²⁻ and manganate(VII) ions MnO₄⁻, the ethanedioate solution needs to be at about 60 °C at the start.

However, during the titration, as FA 1 is added, the temperature of the mixture decreases. This decrease in temperature did not cause the rate of reaction between C₂O₄²⁻ and MnO₄⁻ added from the burette to decrease.

Suggest an explanation for this.

[1]

The rate of reaction between MnO₄⁻ and C₂O₄²⁻ is slow initially due to the electrostatic repulsion between the two negatively charged ions resulting in high activation energy.

As the reaction proceeds, Mn²⁺, an autocatalyst is generated which helps to increase the rate of the reaction.

M4	
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- (v) Calculate the maximum total percentage uncertainty for your titration in (a)(i) if the associated uncertainty associated with each reading using a 10.0 cm³ pipette and a burette are ± 0.03 cm³ and ± 0.05 cm³ respectively.

$$\% \text{ uncertainty due to pipette} = \frac{\pm 0.03}{10.0} \times 100\% = \underline{\pm 0.3000\%}$$

$$\% \text{ uncertainty due to burette used in titration} = \frac{\pm 0.05 \times 2}{14.80} \times 100\% = \underline{\pm 0.6920\%}$$

$$\text{Maximum total percentage uncertainty} = 0.3000 + 0.6920 = \underline{\pm 0.992\% (3 \text{ s.f.})}$$

Percentage uncertainty = [1]

M5	
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(b) (i) **Titration**

1. Fill the burette with **FA 1**.
2. Pipette 25.0 cm³ of **FA3** into a conical flask and add 25 cm³ of **FA 4** using a measuring cylinder.
3. Titrate **FA 3** with **FA 1** until the end-point is reached.
4. Carry out as many accurate titrations as you think necessary to obtain consistent results. Record your titration results in the space provided below.

	1	2
Final burette reading / cm ³	26.55	35.90
Initial burette reading / cm ³	2.90	12.20
Volume of FA 1 / cm ³	23.65	23.70

✓

✓

- (ii) From your titrations, obtain a suitable volume of **FA 1** to be used in your calculations. Show clearly how you obtained this volume.

[3]

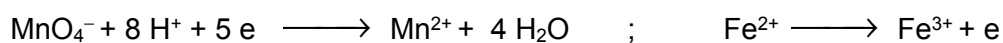
$$\text{Average volume of FA 1 used} = (23.65 + 23.70) \div 2 = \underline{23.68 \text{ cm}^3 (2 \text{ d.p.})}$$

volume of **FA 1** =

M6	M7	M8

(c) Calculations

- (i) The two half-equations for the reaction between the iron(II) salt and manganate(VII) ions are shown below.



Calculate the amount of iron(II) ion present in 1.00 dm³ of **FA 3**.

[1]

$$\text{Amount of KMnO}_4 \text{ in } \underline{23.68} \text{ cm}^3 = \frac{23.68}{1000} \times 0.02108 = \underline{4.992 \times 10^{-4} \text{ mol}}$$



$$\text{Amount of Fe}^{2+} \text{ in } \underline{25.0 \text{ cm}^3} \text{ of FA 3} = 5 \times 4.992 \times 10^{-4} = \underline{2.496 \times 10^{-3} \text{ mol}}$$

$$\text{Amount of Fe}^{2+} \text{ in } \underline{1.00 \text{ dm}^3} \text{ of FA 3} = 2.496 \times 10^{-3} \times \frac{1000}{25.0} = \underline{0.0998 \text{ mol (3 s.f.)}}$$

Amount of iron(II) ions in 1.00 dm³ of **FA 3** =

M9	
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- (ii) Using your answer in (c)(i), determine the value of x , the number of water of crystallisation in the iron(II) salt, $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot x \text{H}_2\text{O}$.

[Ar: Fe, 55.8; H, 1.0; N, 14.0; O, 16.0; S, 32.1]

[3]

$\text{Fe}^{2+} \equiv \text{iron(II) salt} \rightarrow$ Amount of iron(II) salt in 1.00 dm³ of **FA 3** = 0.09983 mol

$$\text{Relative formula mass of iron(II) salt} = \frac{38.4}{0.09983} = 384.6$$

$$\text{Formula mass of iron(II) salt} = 2(14.0 + 4.0) + 55.8 + 2(32.1 + 64.0) + x(2.0 + 16.0)$$

$$384.6 = 284.0 + 18x$$

$$100.6 = 18x$$

$$x = 5.59 \approx \underline{6 \text{ (to nearest integer)}}$$

$x =$

M10	M11	M12

- (d) In the titration of the iron(II) salt and manganate(VII) ion in (b)(i), it was observed that the colour of the mixture in the conical flask gradually changes before an excess drop of manganate(VII) turns the mixture pale pink at the end point.

State and explain the colour changes you observed in the titration, in terms of the complex ions involved.

[2]

As the titration proceeds, the colour of the mixture in the conical flask changes from pale green ($[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$) to yellow ($[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$).

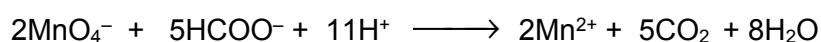
As the electronic configuration of the metal changes, the amount of splitting of the d orbitals (energy gap, ΔE) also changes, resulting in different wavelength of light being absorbed during the d-d transitions.

M13	M14

(e) **Planning**

A saturated aqueous solution of magnesium methanoate, $\text{Mg}(\text{HCOO})_2$, has a solubility of approximately **150 g dm⁻³** at room temperature. Its exact solubility can be determined by titrating magnesium methanoate against aqueous potassium manganate(VII).

During the titration, methanoate ion, HCOO^- reacts with manganate(VII) ion, MnO_4^- in an acidic medium as shown in the equation below.



You may assume that you are provided with:

- 100 g of solid magnesium methanoate, $\text{Mg}(\text{HCOO})_2$
- 0.0200 mol dm⁻³ aqueous potassium manganate(VII), KMnO_4
- 1.00 mol dm⁻³ sulfuric acid, H_2SO_4
- the apparatus and equipment normally found in a school or college laboratory.

In order to obtain a reliable and feasible titre value, the saturated solution of magnesium methanoate needs to be diluted accurately (~**50 times** dilution) before titration.

- (i) Plan an investigation to determine the effect of temperature on the solubility of magnesium methanoate, $\text{Mg}(\text{HCOO})_2$. Your plan must enable you to plot a graph that includes data from the experiment.

Your plan should include details of:

- the preparation of 50 cm³ of **saturated** aqueous solutions of Mg(HCOO)₂ at different temperatures
- the preparation of diluted solutions of Mg(HCOO)₂ for titration
- an outline of how the titration could be carried out
- a sketch of the graph you would expect to obtain
- an outline how the results can be used to determine the effect of temperature on the solubility of Mg(HCOO)₂

[6]

YOU ARE NOT REQUIRED TO PERFORM THIS EXPERIMENT.

Preparation of saturated solution of HCOO⁻

- 1) Weigh approximately 8.0 g of solid Mg(HCOO)₂ and transfer the solid to 50 cm³ of water in a beaker. Stir to ensure even mixing such that the obtained solution is saturated. Ensure solid deposits can be seen in the beaker.
- 2) Place this beaker in a thermostatically controlled water bath set at 30°C.
- 3) Use a thermometer to measure the temperature of the solution in the beaker.-Record the temperature of the solution.

Preparation of diluted solution

- 4) Quickly decant the saturated solution into a burette.
- 5) Using the burette, dispense 5.00 cm³ of the saturated HCOO⁻ solution into a 250 cm³ volumetric flask.
- 6) Top up to the mark with deionized water, and invert the flask to ensure even mixing to obtain a homogenous solution. Label this as diluted solution at 30°C.

Titration

- 7) Fill another burette with KMnO₄ (aq)
- 8) Pipette 25.0 cm³ of the diluted HCOO⁻ solution into a conical flask and add 20 cm³ of dilute H₂SO₄ using a measuring cylinder.
- 9) Note the initial reading of the burette.
- 10) Titrate with KMnO₄ until a permanent pale pink colour is obtained. Note the final reading of the burette.
- 11) Carry out accurate titrations to obtain consistent results. Record your titration results.

Repeat Step 1 to 11 at different temperatures (i.e. 35°C, 40°C, 45°C, 50°C) by adjusting the temperature of the thermostatically controlled water bath.

Analysis of Results

Determine the average titre value of KMnO₄ used and determine the amount of MnO₄⁻ used, which can be used to determine the amount of HCOO⁻ in the diluted solution and hence the concentration of HCOO⁻ in the diluted and saturated solution. This can be correlated to determine the solubility of Mg(HCOO)₂.

Presentation of sample calculations

$$\text{Amount of MnO}_4^- = \frac{\text{Volume of KMnO}_4 \text{ used}}{1000} \times 0.0200$$



$$\text{Amount of HCOO}^- \text{ in } 25.0 \text{ cm}^3 \text{ (diluted solution)} = \frac{5}{2} \times \frac{\text{Volume of KMnO}_4 \text{ used}}{1000} \times 0.0200$$

$$\text{Amount of HCOO}^- \text{ in } 5.00 \text{ cm}^3 \text{ (saturated)} = \frac{250}{25.0} \times \frac{5}{2} \times \frac{\text{Volume of KMnO}_4 \text{ used}}{1000} \times 0.0200$$

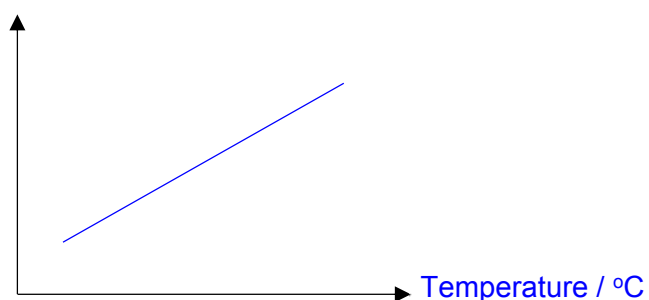
$$[\text{HCOO}^-] \text{ (saturated)} = \frac{\frac{250}{25.0} \times \frac{5}{2} \times \frac{\text{Volume of KMnO}_4 \text{ used}}{1000} \times 0.0200}{\left(\frac{5.00}{1000} \right)}$$

$$[\text{Mg}(\text{HCOO})_2] \text{ (saturated)} = \frac{\frac{250}{25.0} \times \frac{5}{2} \times \frac{\text{Volume of KMnO}_4 \text{ used}}{1000} \times 0.0200}{\left(\frac{5.00}{1000} \right) \times 2}$$

$$\text{Solubility of Mg}(\text{HCOO})_2 \text{ (saturated)} = \frac{\frac{250}{25.0} \times \frac{5}{2} \times \frac{\text{Volume of KMnO}_4 \text{ used}}{1000} \times 0.0200}{\left(\frac{5.00}{1000} \right) \times 2} \times 114.3$$

Repeat the above for each different temperature. Plot a graph of solubility of $\text{Mg}(\text{HCOO})_2$ against temperature to see how temperature affects the solubility of $\text{Mg}(\text{HCOO})_2$.

Solubility of
 $\text{Mg}(\text{HCOO})_2 / \text{g dm}^{-3}$



M15 – M20

- (ii) A student used the same titration method, this time to measure the concentration of a saturated solution of barium methanoate. Explain why the acidification of the solution with dilute sulfuric acid might make the titration difficult to do.

[1]

A precipitate of BaSO₄ will be formed thus obstructing an accurate judgement of the end-point of the titration.

M21	
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[Total: 21]

- 2 Strong acids, such as hydrochloric acid, HCl, are completely ionised in aqueous solution. Weak acids, such as ethanoic acid, CH₃COOH, are partially ionised in aqueous solution.

You will investigate the enthalpy change for the reaction of an excess of each of these acids with magnesium and hence determine the energy needed to cause the weak acid to ionise completely.

(a) **Experiment 1**

FA 5 is 1.0 mol dm⁻³ hydrochloric acid, HCl.

FA 6 is magnesium, Mg.

1. Weigh the magnesium strip, **FA 6**, and record the mass reading.
2. Support the Sytrofoam cup in the 250 cm³ beaker.
3. Coil the magnesium strip loosely so that it fits into the bottom of the Sytrofoam cup and then remove the strip.
4. Use the measuring cylinder to transfer 25 cm³ of the acid, **FA 5**, into the Sytrofoam cup.
5. Place the thermometer in the acid and read the initial temperature. This is the temperature at time zero ($t = 0$).
6. Start timing and do not stop the stop watch until the whole experiment has been completed.
7. Read the temperature of the acid every half minute for two minutes.
8. At time $t = 2.5$ minutes drop the magnesium strip, **FA 6**, into the acid and stir the mixture. **Do not try to read the temperature at this time.**
9. Measure and record, the temperature of the mixture at $t = 3.0$ minutes and then every half minute until $t = 8.0$ minutes. Stir the mixture continuously between thermometer readings.
10. Rinse the Sytrofoam cup for use in **Experiment 2**. Shake to remove excess water.

In the space provided below, record all measurements of mass. Record all your results of temperature and time in a table.

Results

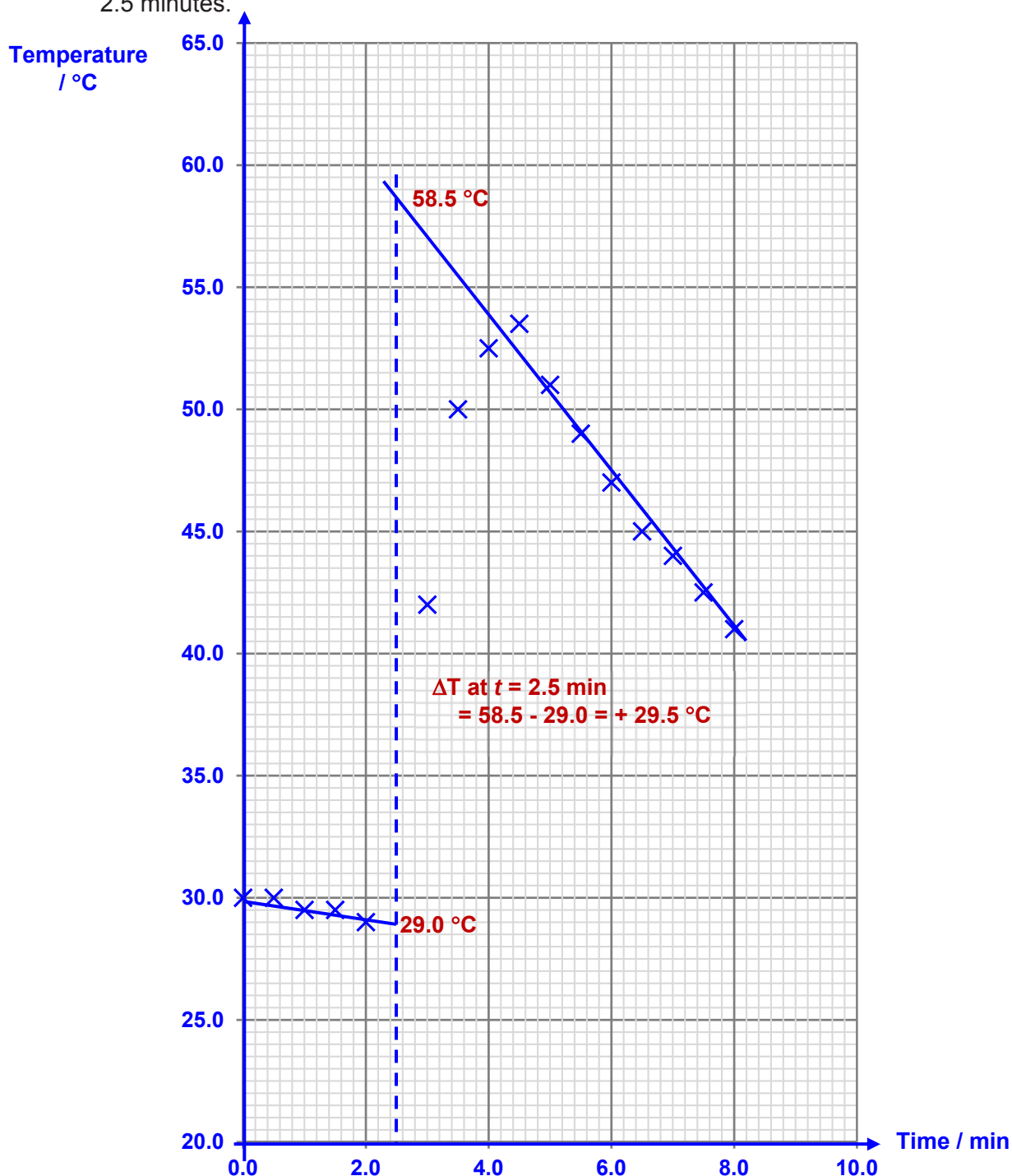
Mass of magnesium strip = 0.162 g

Time / min	Temperature / °C	Time / min	Temperature / °C
0.0	30.0	5.5	49.0
0.5	30.0	6.0	47.0
1.0	29.5	6.5	45.0
1.5	29.5	7.0	44.0
2.0	29.0	7.5	42.5
2.5	—	8.0	41.0
3.0	42.0		
3.5	50.0		
4.0	52.5		
4.5	53.5		
5.0	51.0		

[4]

M1	M2	M3	M4

- (b) Plot a graph of temperature on the y-axis against time on the x-axis on the grid below. The scale for the temperature axis should extend 10 °C greater than the maximum temperature you recorded. You will use the graph to determine the theoretical maximum temperature rise at 2.5 minutes.



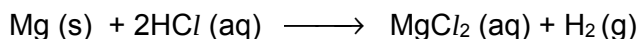
Draw two lines of best fit through the points on your graph, the first for the temperature before adding FA 6 and the second for the cooling of the mixture once the reaction is complete.

Extrapolate the two lines to 2.5 minutes, draw a vertical line between the two and determine the theoretical rise in temperature at this time.

theoretical rise in temperature at 2.5 minutes = [4]

M5	M6	M7	M8

- (c) Magnesium reacts with hydrochloric acid according to the equation shown.



- (i) Use your answer to (b) to calculate the heat energy, in joules, given out when **FA 6** is added to the acid.

[Assume 4.2 J of heat energy raises the temperature of 1.0 cm³ of the mixture by 1.0 °C.]

[1]

$$\begin{aligned} \text{Energy evolved} &= 25.0 \times 4.2 \times \Delta T \\ &= 25.0 \times 4.2 \times (58.5 - 29.0) \\ &= 3098 \text{ J} \end{aligned}$$

heat energy evolved =

M9	
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- (ii) Calculate the enthalpy change when 1 mole of **FA 6**, Mg, reacts with hydrochloric acid.
[Ar: Mg, 24.3]

[1]

$$\Delta H = -\frac{Q}{n_{\text{limiting}}} = -\frac{3098}{\frac{0.158}{24.3}} = -476.3 \text{ kJ mol}^{-1}$$

enthalpy change, ΔH =

M10	
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- (d) (i) Calculate the percentage error in the measurement of ΔT in the experiment.

[1]

$$\text{Percentage error} = \left(\frac{\pm 0.5 \times 2}{29.5} \right) \times 100\% = \pm 3.39 \%$$

M11	
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- (ii) A student suggested that the experiment carried out in (a) could be improved by using a catalyst.

Would the use of a catalyst improve the accuracy of the results in this experiment? Give a reason for your answer.

[1]

YES because the reaction is faster so less heat is lost. Hence, ΔT will be more accurately determined.

OR NO because a catalyst does not alter ΔH or ΔT

M12	
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- (ii) Another student could not find the hydrochloric acid, **FA 5**, and used sulfuric acid, H_2SO_4 , instead. He used the same volume and the same concentration as the hydrochloric acid in **FA 5**.

What effect would this change have on the temperature rise in the experiment? Give a reason for your answer.

[1]

No effect because the acid is in excess / magnesium is the limiting reagent /

all the Mg has reacted.

OR

ΔT would be larger because the reaction is faster as acid is diprotic / dibasic so less

heat is lost to surroundings

M13	
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(e) **Experiment 2**

FA 7 is 1.0 mol dm^{-3} ethanoic acid, CH_3COOH

FA 8 contains one strip of magnesium, Mg of a mass of 0.087 g

Read the whole **Experiment 2** before starting any practical work.

1. Support the Sytrofoam cup in the 250 cm^3 beaker.
2. Coil the magnesium strip loosely so that it fits into the bottom of the Sytrofoam cup and then remove the strip.
3. Use the measuring cylinder to transfer 25 cm^3 of the acid, **FA 7**, into the Sytrofoam cup.
4. Place the thermometer in the acid and measure and record the initial temperature of the acid.
5. Add the piece of magnesium, **FA 8**, into the acid in the cup.
6. Using the thermometer, stir the mixture continuously until it reaches its maximum temperature. This could take several minutes. Record this temperature, T_{max} .
7. Calculate and record the change in temperature, ΔT .
8. Complete **Table 2.1** by recording all measurements of temperature.

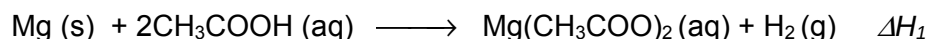
Table 2.1

Initial temperature / $^{\circ}\text{C}$	31.0
Maximum temperature, T_{max} / $^{\circ}\text{C}$	45.0
Change in temperature, ΔT / $^{\circ}\text{C}$	+14.0

[1]

M14	
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Magnesium reacts with ethanoic acid according to the equation shown.



Use your results in Table 2.1 to calculate the enthalpy change, ΔH_1 , when 1 mole of Mg, FA 8, reacts with excess ethanoic acid. Assume 4.2 J of heat energy changes the temperature of 1.0 cm³ of the mixture by 1.0 °C. [A_r: Mg, 24.3]

[1]

Amount of heat energy evolved, Q = $m\Delta T$ = 25.0 x 4.2 x 14.0 = **1470 J**

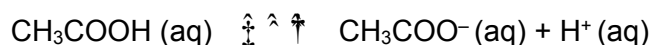
$$\Delta H_1 = -\frac{Q}{n_{\text{limiting}}} = -\frac{1470}{\frac{0.087}{24.3}} = -410.6 \text{ kJ mol}^{-1}$$

$$\Delta H_1 = \dots\dots\dots$$

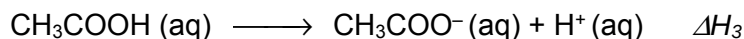
M15	
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- (f) Hydrochloric acid is a strong acid; it is fully ionised in aqueous solution. When the same experiment in (e) was repeated with hydrochloric acid, **FA 5**, instead of ethanoic acid, the enthalpy change, ΔH_2 , when 1 mole of Mg reacts with excess hydrochloric acid was determined to be $-461.1 \text{ kJ mol}^{-1}$.

Ethanoic acid is a weak acid. It is partially ionised in aqueous solution.



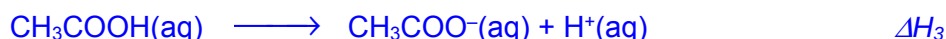
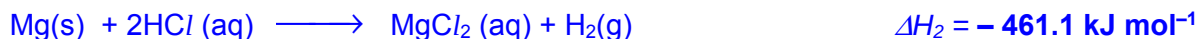
You are to determine the energy needed for one mole of ethanoic acid to ionise completely.



Use the values for the enthalpy changes, ΔH_1 you obtained in (e) and the ΔH_2 provided to calculate the energy change for the ionisation of ethanoic acid, ΔH_3 .

Show clearly how you obtained your answer.

[2]



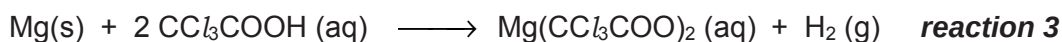
$$\Delta H_3 = \frac{1}{2}(\Delta H_1 - \Delta H_2) = \frac{1}{2}(-410.6 - (-461.1))$$

$$= +25.3 \text{ kJ mol}^{-1}$$

$$\Delta H_3 = \dots\dots\dots$$

M16	M17
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- (g) The experiment in (e) was repeated using trichloroethanoic acid instead of ethanoic acid.



Trichloroethanoic acid, CCl_3COOH , is a weak acid that is however stronger than ethanoic acid. The enthalpy change for **reaction 3** is between the two values given in **Table 2.2**.

Table 2.2

reaction	equation	$\Delta H / \text{kJ mol}^{-1}$
1	$\text{Mg (s)} + 2\text{CH}_3\text{COOH (aq)} \longrightarrow \text{Mg(CH}_3\text{COO)}_2 \text{ (aq)} + \text{H}_2 \text{ (g)}$	-460.3
2	$\text{Mg (s)} + 2\text{HCl (aq)} \longrightarrow \text{MgCl}_2 \text{ (aq)} + \text{H}_2 \text{ (g)}$	-464.1

- (i) Explain why the enthalpy change for reaction 3 is more exothermic than the enthalpy change for reaction 1.

[1]

CCl_3COOH is a stronger acid than ethanoic acid.

Increased electronegativity of Cl / electron withdrawing Cl group increases the strength of (trichloroethanoic) acid / makes it easier to release H^+ / deprotonate/ dissociate / ionise. (compared to ethanoic acid). Hence, less energy is required to dissociate H^+ from CCl_3COOH compared to CH_3COOH and reaction 3 is more exothermic.

M18

- (ii) Explain why the enthalpy change for reaction 3 is less exothermic than the enthalpy change for reaction 2.

[1]

Cl_3COOH is a weaker acid than HCl .

Hence, some energy released during neutralisation is used to dissociate Cl_3COOH completely.

M19

[Total: 19]

3 Qualitative Analysis

At each stage of any test you are to record details of the following.

- colour changes seen
- the formation of any precipitate
- the solubility of such precipitates in an excess of the reagent added

Where reagents are selected for use in a test, the **name** or **correct formula** of the element or compound must be given.

Where gases are released they should be identified by a test, **described in the appropriate place in your observations**.

You should indicate clearly at what stage in a test a change occurs.

No additional tests for ions present should be attempted.

(a) (i) **FA 9, FA 10 and FA 11** each contain one anion and one cation.

Carry out the following tests and record your observations.

Test	observations		
	FA 9	FA 10	FA 11
To a 1 cm depth of solution in a test-tube, add a few drops of aqueous silver nitrate, then	• White ppt formed.	• No ppt formed.	• No ppt formed.
add aqueous ammonia.	• White ppt is soluble in excess $\text{NH}_3(\text{aq})$ to form a colourless solution.	• No ppt formed.	• No ppt formed.
To a 1 cm depth of solution in a test-tube, add a few drops of aqueous barium nitrate, then	• No ppt formed.	• White ppt formed.	• No ppt formed.
add dilute nitric acid.	• No ppt formed.	• White ppt is insoluble in excess $\text{HNO}_3(\text{aq})$.	• No ppt formed.
To a 1 cm depth of solution in a test-tube, add a spatula of solid sodium carbonate.	• Effervescence seen. • Gas evolved formed a white ppt with limewater.	• Effervescence seen. • Gas evolved formed a white ppt with limewater.	• Effervescence seen. • Gas evolved formed a white ppt with limewater.

M1	M2	M3	M4

- (ii) What cation is present in **FA 9**, **FA 10** and **FA 11**?

H^+ / hydrogen ions

- (iii) Suggest another chemical test that you could carry out to confirm the presence of the cation you identified in (ii).

Carry out this test on one of **FA 9**, **FA 10** and **FA 11** and record your observations.

Test: Add some Mg turnings to 2 cm depth of **FA 9** / **FA 10** / **FA 11**.

Observations: Effervescence seen Gas extinguishes a lighted splint with a 'pop' sound

- (iv) Complete the table to identify, as far as possible, the anions present in **FA 9**, **FA 10** and **FA 11**. If you are not able to identify the anion from the tests you carried out in (i), write 'unknown'.

	FA 9	FA 10	FA 11
ion present	Cl^-	SO_4^{2-}	unknown

M5	M6	M7	M8

- (v) For any one anion that you were unable to identify in (iv) you are to devise a test or tests that will enable you to identify it. You can assume that it is one of the anions listed in the Qualitative Analysis Notes.

Carry out the test(s), record the observation(s) you obtained and identify the unknown anion. In the space below, draw a single table to record details of the test(s) performed and observations made.

No.	Test	Observation
1	Add 1 cm depth of FA 11 into the test-tube. Add 1 cm depth of NaOH(aq). Then add aluminium foil to the mixture and warm gently.	Effervescence observed upon heating with Al. Gas evolved turns moist red litmus blue.
2	Place 1 cm depth of FA 11 in a test-tube. Add 1 cm depth of hydrochloric acid solution to FA 11. Warm. *	No brown fumes evolved

Identify the anion and state the evidences for your identification.

Anion: NO_3^-

Evidence: In test (1), NH_3 (g) was evolved on heating with aq NaOH and Al, which turns moist red litmus paper blue. No brown NO_2 gas was evolved with aq HCl in test (2).

[10]

M9	M10

- (b) **FA 12** is an aqueous solution of a mixture containing two anions. Carry out the following tests and record your observations.

Test	Observations
To a 1 cm depth of FA 12 in a test-tube, add a 1 cm depth of dilute hydrochloric acid, then	<u>Effervescence</u> seen. <u>Gas evolved</u> formed a <u>white ppt with limewater</u>.
add a few drops of hydrogen peroxide, then	<u>Brown / yellow / red- brown / yellow-brown solution</u> formed
add a few drops of starch.	<u>Blue-black / black / dark blue solution</u> formed.
To a 1 cm depth of FA 12 in a test-tube, add a 3 cm depth of aqueous copper(II) sulfate, then	<u>Blue / green / brown ppt</u> observed
add a 1 cm depth of dilute hydrochloric acid, then	<u>Brown solution</u> formed
add aqueous sodium thiosulfate.	<u>White / cream / off-white / pale grey and solid / residue / ppt</u>

From these observations, identify two ions present in **FA 12**.

ions present in **FA 12** CO_3^{2-} and I^-

[5]

M11	M12	M13	M14	M15

[Total: 15]